

DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO

Test Booklet No. :

Series

05821

TEST BOOKLET

CIVIL ENGINEERING

(Paper—I)



Time Allowed : 2 Hours

Full Marks : 100

Read the following instructions carefully before you begin to answer the questions :

1. The name of the Subject, Roll Number as mentioned in the Admission Certificate, Test Booklet No. and Series are to be written legibly and correctly in the space provided on the Answer-Sheet with Black/Blue ballpoint pen.
2. **Answer-Sheet without marking Series as mentioned above in the space provided for in the Answer-Sheet shall not be evaluated.**
3. All questions carry equal marks.

The Answer-Sheet should be submitted to the Invigilator.

Directions for giving the answers : Directions for answering questions have already been issued to the respective candidates in the 'Instructions for marking in the OMR Answer-Sheet' along with the Admit Card and Specimen Copy of the OMR Answer-Sheet.

Example :

Suppose the following question is asked :

The capital of Bangladesh is

- (A) Chennai
- (B) London
- (C) Dhaka
- (D) Dhubri

You will have four alternatives in the Answer-Sheet for your response corresponding to each question of the Test Booklet as below :

(A) (B) (C) (D)

In the above illustration, if your chosen response is alternative (C), i.e., Dhaka, then the same should be marked on the Answer-Sheet by blackening the relevant circle with a Black/Blue ballpoint pen only as below :

(A) (B) (C) (D)

The example shown above is the only correct method of answering.

4. Use of eraser, blade, chemical whitener fluid to rectify any response is prohibited.
5. Please ensure that the Test Booklet has the required number of pages (16) and 100 questions immediately after opening the Booklet. In case of any discrepancy, please report the same to the Invigilator.
6. No candidate shall be admitted to the Examination Hall/Room 20 minutes after the commencement of the examination.
7. No candidate shall leave the Examination Hall/Room without prior permission of the Supervisor/Invigilator. No candidate shall be permitted to hand over his/her Answer-Sheet and leave the Examination Hall/Room before expiry of the full time allotted for each paper.
8. No Mobile Phone, Electronic Communication Device, etc., are allowed to be carried inside the Examination Hall/Room by the candidates. Any Mobile Phone, Electronic Communication Device, etc., found in possession of the candidate inside the Examination Hall/Room, even if on off mode, shall be liable for confiscation.
9. No candidate shall have in his/her possession inside the Examination Hall/Room any book, notebook or loose paper, except his/her Admission Certificate and other connected papers permitted by the Commission.
10. Complete silence must be observed in the Examination Hall/Room. No candidate shall copy from the paper of any other candidate, or permit his/her own paper to be copied, or give, or attempt to give, or obtain, or attempt to obtain irregular assistance of any kind.
11. This Test Booklet can be carried with you after answering the questions in the prescribed Answer-Sheet.
12. Noncompliance with any of the above instructions will render a candidate liable to penalty as may be deemed fit.
13. No rough work is to be done on the OMR Answer-Sheet. You can do the rough work on the space provided in the Test Booklet.

N.B. : There will be negative marking @ 0.25 per 1 (one) mark against each wrong answer.

1. If the size of a column is reduced above the floors, the main bar of the column
 - (A) continues up
 - (B) bends inward at the floor level
 - (C) stops just below the floor level and separates lap bars provided
 - (D) All of the above
2. The stone suitable for damp-proofing is
 - (A) slate
 - (B) marble
 - (C) laterite
 - (D) granite
3. For constructing road pavements, the type of cement generally used is
 - (A) ordinary Portland cement
 - (B) rapid hardening cement
 - (C) low heat cement
 - (D) blast furnace slag cement
4. Workability of concrete is directly proportional to
 - I. time of transit
 - II. water-cement ratio
 - III. grading of aggregate
 - IV. strength of aggregate
 - V. aggregate-cement ratio

Select the correct answer using the codes given below.

 - (A) III, IV and V
 - (B) I, II and IV
 - (C) II, III and V
 - (D) II and III
5. Calculate the quantity of metal required for 3.7 m wide macadam road for 1 km length for 1 layer of 8 cm compacted thickness.
 - (A) 444 cu m
 - (B) 565 cu m
 - (C) 454 cu m
 - (D) 765 cu m
6. If the intercept on vertical staff is observed as 1.5 m from a tacheometer, the horizontal distance between the tacheometer and staff station is
 - (A) 75 m
 - (B) 100 m
 - (C) 150 m
 - (D) 300 m
7. Groundwater is generally free from
 - (A) suspended impurities
 - (B) dissolved impurities
 - (C) Both (A) and (B)
 - (D) None of the above
8. Coarse sand has a fineness modulus in the range of
 - (A) 2.2-2.4
 - (B) 2.4-2.6
 - (C) 2.6-2.9
 - (D) 2.9-3.2

9. Black cotton soil is not suitable for foundation because of its
- (A) low bearing capacity
 - (B) cohesive particles
 - (C) swelling and shrinkage
 - (D) black colour
10. High percentage of C_3S and low percentage of C_2S in a cement will result in
- I. rapid hardening
 - II. high early strength with high heat generation
 - III. more resistance to chemical attack
- Select the correct answer using the codes given below.
- (A) Only I
 - (B) Only III
 - (C) Both I and II
 - (D) Both I and III
11. Calculate the area (in m^2) of the formwork required for a beam of 2 m span and cross-section dimension of 400 mm $\times$ 200 mm.
- (A) 0.8
 - (B) 0.16
 - (C) 1.2
 - (D) 2
12. For one cubic metre of concrete (1:2:4), the number of cement bags required is
- (A) 4.5
 - (B) 5.0
 - (C) 5.3
 - (D) 6.3
13. The number of bars to be spliced at a section should not exceed
- (A) 50%
 - (B) 40%
 - (C) 25%
 - (D) 30%
14. The process of adding strength to a weak or damaged building is known as
- (A) repair
 - (B) retrofitting
 - (C) restoration
 - (D) reconstruction
15. Excess of silica in clay
- (A) makes the brick brittle and weak
 - (B) makes the brick crack and warp on drying
 - (C) changes the colour of brick from red to yellow
 - (D) improves impermeability and durability of the brick

16. The size of ballast used in Indian Railways for wooden sleepers is
- (A) 25 mm
 - (B) 38 mm
 - (C) 43 mm
 - (D) 50 mm
17. As per Indian Standard, the length of one link in 30 m chain is
- (A) 20 cm
 - (B) 30 cm
 - (C) 40 cm
 - (D) 10 cm
18. Which of the following is an activity?
- (A) Construction of foundation
 - (B) Construction of roof
 - (C) Construction of superstructure
 - (D) All of the above
19. Minimum thickness of wall where single flemish bond can be used is
- (A) half-brick thick
 - (B) one-brick thick
 - (C) one and a half-brick thick
 - (D) two-brick thick
20. What is the value of camber that should be provided in case of WBM pavement surface in an area of heavy rainfall?
- (A) 1 in 30
 - (B) 1 in 48
 - (C) 1 in 60
 - (D) 1 in 72
21. If P is the standard consistency of the given cement sample, then the quantity of water to be added to prepare a cement paste for determining the initial setting time is
- (A) $\left(\frac{P}{4} + 3\right)\%$ of weight of cement
 - (B) 0.85% of weight of cement
 - (C) $P\%$ of weight of cement
 - (D) 0.78% of weight of cement
22. The digested sludge from septic tank is removed after a maximum period of
- (A) 3 years
 - (B) 3.5 years
 - (C) 4 years
 - (D) 4.5 years

23. The DPC is provided for efficiency at

- (A) just below the ground level
- (B) the window sill level
- (C) above the lintel
- (D) anywhere in the wall

24. The operation of finishing mortar joint after completion of masonry wall is called

- (A) toothing
- (B) pointing
- (C) racking back
- (D) jointing

25. The metric scale recommended by BIS for the use of technical drawing in case of full scale is

- (A) 1 : 1
- (B) 1 : 2
- (C) 5 : 1
- (D) 1 : 5

26. Interception loss is

- (A) more towards the end of the storm
- (B) more at the beginning of the storm
- (C) uniform throughout the storm
- (D) low in the beginning of the storm and gradually increases

27. The length of a transition curve is proportional to

- (A) $\frac{V}{R}$
- (B) $\frac{V^2}{R}$
- (C) $\frac{V^3}{R}$
- (D) $\frac{V^4}{R}$

(where V = velocity and R = radius of curvature)

28. Level crossing is a/an

- (A) regulatory sign
- (B) warning sign
- (C) informatory sign
- (D) None of the above

29. A 10% flood frequency means

- (A) flood will increase by 10% every year
- (B) every tenth year there will be flood
- (C) flood has 10 out of 100 chances of being equalled or exceeded
- (D) Any one of the above

30. When a slab is continuous over several spans, negative (i.e., hogging) bending moment is induced over the
- (A) end support
 - (B) intermediate support
 - (C) Both (A) and (B)
 - (D) None of the above
31. Dechlorination of water is achieved by adding
- (A) sodium thiosulphate
 - (B) sodium sulphate
 - (C) sodium bisulphate
 - (D) None of the above
32. The coagulant widely used for sewage treatment is
- (A) alum
 - (B) ferric chloride
 - (C) ferric sulphate
 - (D) chlorinated copperas
33. The water stored in the reservoir below the minimum pool level is called
- (A) useful storage
 - (B) dead storage
 - (C) valley storage
 - (D) surcharge storage
34. The percentage of reinforcement in case of slabs when high-strength deformed bars are used is not less than
- (A) 0.15
 - (B) 0.12
 - (C) 0.30
 - (D) 1.00
35. As per IS 383, sand can be graded into _____ zones.
- (A) 2
 - (B) 5
 - (C) 3
 - (D) 4
36. If longitudinal strain is tensile, the lateral strain will be
- (A) tensile
 - (B) compressive
 - (C) zero
 - (D) Not determined
37. The minimum and maximum percentages of longitudinal reinforcement in column according to IS 456-2000 are _____ respectively.
- (A) 0.8% and 0.3%
 - (B) 1% and 8%
 - (C) 0.8% and 4%
 - (D) 0.8% and 6%

38. The variation of the bending moment in the portion of a beam carrying linearly varying load is
- (A) linear
 - (B) parabolic
 - (C) cubic
 - (D) constant
39. Elementary profile of a gravity dam is
- (A) a trapezoidal figure
 - (B) a pyramid
 - (C) a right-angled triangle
 - (D) a square
40. Effective length of a column is the length between the points of
- (A) maximum momentum
 - (B) zero shear
 - (C) zero moment
 - (D) None of the above
41. The standard project flood is
- (A) the same as maximum probable flood
 - (B) the same as designed flood
 - (C) smaller than the maximum probable flood
 - (D) larger than the maximum probable flood by a factor implying safety factor
42. Sluices are provided in a dam to
- (A) supply water to downstream area
 - (B) reduce flood level
 - (C) remove sediment from reservoirs
 - (D) All of the above
43. Base flow is
- (A) the annual minimum flow in a stream
 - (B) the flow not obstructed by man-made structure
 - (C) delayed groundwater flow reaching a stream
 - (D) None of the above
44. Chemical weathering of soil is caused due to
- (A) oxidation
 - (B) hydration
 - (C) carbonation and leaching
 - (D) All of the above
45. The minimum width of septic tank is taken as
- (A) 70 cm
 - (B) 75 cm
 - (C) 80 cm
 - (D) 90 cm

46. The method of shoring adopted in loose soil when depth is not excess of 4 m is
- (A) stray bracing
 - (B) box sheeting
 - (C) vertical sheeting
 - (D) runner system
47. Le Chatelier's method can be used to determine
- (A) fineness of cement
 - (B) fineness of aggregate
 - (C) soundness of cement
 - (D) compressive strength of cement
48. The ratio of flexural rigidity of a beam ($b \times d$) to another ($b \times 2d$) of similar material will be
- (A) $\frac{1}{2}$
 - (B) $\frac{1}{4}$
 - (C) $\frac{1}{8}$
 - (D) $\frac{1}{18}$
49. In order to prevent segregation in concrete, dropping height should not be more than
- (A) 25 cm
 - (B) 50 cm
 - (C) 100 cm
 - (D) 150 cm
50. BOD test is conducted at
- (A) ambient temperature
 - (B) a temperature of 15 °C
 - (C) a temperature of 20 °C
 - (D) a temperature of 27 °C
51. If the pores of a soil are completely full of air only, the soil is said to be
- (A) wet soil
 - (B) fully saturated soil
 - (C) dry soil
 - (D) partially saturated soil
52. The forebearings of the lines AB and BC are 40° and 120° respectively. The included angle between AB and BC is
- (A) 40°
 - (B) 60°
 - (C) 80°
 - (D) 100°
53. Workability of concrete for a given water content is good if the aggregate is
- (A) angular aggregate
 - (B) flaky aggregate
 - (C) rounded aggregate
 - (D) irregular aggregate

54. Moment of inertia of a circular section about its diameter d is

- (A) $\frac{\pi d^3}{16}$
- (B) $\frac{\pi d^4}{32}$
- (C) $\frac{\pi d^3}{32}$
- (D) $\frac{\pi d^4}{64}$

55. Mercury is used in barometers because

- (A) it is a perfect fluid
- (B) its volume changes with temperature
- (C) it is a liquid metal
- (D) it gives less height of column for high pressure

56. A surge tank is provided in the hydropower scheme to

- (A) strengthen the penstocks
- (B) reduce water hammer pressure
- (C) reduce frictional losses in the system
- (D) minimize the cost of hydro-power development

57. A self-recording rain gauge

- (A) records the cloud cover
- (B) records the snow melt
- (C) records the rainfall intensity
- (D) records the cumulative depth of rainfall

58. The plan of a building is drawn by assuming the structure to be cut at

- (A) plinth level
- (B) window sill level
- (C) ground level
- (D) top of the door level

59. DAD (Depth Area Duration) studies for a particular storm indicate that the average depth of rainfall

- (A) remains constant
- (B) decreases as the area decreases
- (C) decreases as the area increases
- (D) increases as the duration of rainfall increases

60. The river which builds up its bed progressively is called

- (A) aggrading river
- (B) degrading river
- (C) boulder river
- (D) deltaic river

61. Tensile strength of wood is

- (A) maximum in the direction parallel to the grains
- (B) minimum in the direction parallel to the grains
- (C) maximum in the direction across the grains
- (D) same in all directions

62. For the design of superelevation for mixed traffic conditions, the speed is reduced by
- (A) 15%
 - (B) 20%
 - (C) 25%
 - (D) 75%
63. Intersection method of detailed plotting is most suitable for
- (A) forests
 - (B) urban areas
 - (C) hilly areas
 - (D) plains
64. When the magnetic bearing of the sun at noon is $185^{\circ}20'$, the magnetic declination will be
- (A) $5^{\circ}20'$ East
 - (B) $5^{\circ}20'$ West
 - (C) $5^{\circ}20'$ North
 - (D) $5^{\circ}20'$ South
65. Wind-blown silt having little or no stratification is called
- (A) talus
 - (B) drift
 - (C) peat
 - (D) loess
66. The type of valve, which is provided on the suction pipe in a tube-well is
- (A) air relief valve
 - (B) reflux valve
 - (C) pressure relief valve
 - (D) sluice valve
67. An advantage of COD test over BOD test is
- (A) it oxidizes all matters including toxins
 - (B) it cannot oxidize benzene, pyridine, etc.
 - (C) it is relatively a quick process
 - (D) biodegradable matter cannot be distinguished from the rest
68. Lead acetate test in sewer manhole is done to test the presence of
- (A) methane gas
 - (B) hydrogen sulphide
 - (C) carbon dioxide gas
 - (D) diesel vapours
69. A milestone chart
- (A) depicts the delay of job
 - (B) shows the interdependence of various jobs
 - (C) shows the events in chronological, but not in a logical sequence
 - (D) All of the above

70. Which is **not** a method of controlled flooding in irrigation methods?
- (A) Contour
 - (B) Ring basin
 - (C) Check basin
 - (D) Border strip
71. Which of the following structures is constructed to separate under sluices from the main weir?
- (A) Marginal bund
 - (B) Divide wall
 - (C) Head regulator
 - (D) None of the above
72. A wooden block hinged on post outside a door, is known as
- (A) cleat
 - (B) stop
 - (C) horn
 - (D) None of the above
73. Which of the following lines are used to show that the object is cut and then viewed?
- (A) Hidden lines
 - (B) Leader lines
 - (C) Center lines
 - (D) Hatching lines
74. First angle and third angle projection methods are used in drawing
- (A) perspective projection
 - (B) oblique projection
 - (C) areal projection
 - (D) orthographic projection
75. A joint between two components lying approximately in the same plane is called
- (A) corner joint
 - (B) T-joint
 - (C) lap joint
 - (D) butt joint
76. The neutral axis of an over-reinforced section falls
- (A) on the critical neutral axis of balanced section
 - (B) below the critical neutral axis of balanced section
 - (C) above the critical neutral axis of balanced section
 - (D) None of the above
77. A floating body is said to be in a state of stable equilibrium
- (A) when its metacentric height is zero
 - (B) when the metacentre is above the centre of gravity
 - (C) when the metacentre is below the centre of gravity
 - (D) only when its centre of gravity is below its centre of buoyancy

78. Which of the following combinations for a channel is most suitable to achieve best results in flood management?

- (A) Increasing width and roughness
- (B) Increasing width and decreasing roughness
- (C) Decreasing width and increasing roughness
- (D) Decreasing width and roughness

79. Bar charts are suitable for

- (A) minor works.
- (B) major works
- (C) large projects
- (D) All of the above

80. Which of the following represents a metamorphic rock?

- I. Slate
- II. Shale
- III. Quartzite

Select the correct answer using the codes given below.

- (A) Only III
- (B) Both I and III
- (C) Both II and III
- (D) All I, II and III

81. Efflorescence test is conducted for burnt clay bricks to find out the

- (A) presence of alkaline substance
- (B) hardness
- (C) soundness
- (D) presence of cracks or holes

82. Which of the following options is **not** a notable challenge while scheduling a project?

- (A) Deadlines exist
- (B) Independent activities
- (C) Too many workers may be required
- (D) Costly delay

83. As per IRC, the slope of earth in cutting should be

- (A) 1 : 2
- (B) 1 : 1
- (C) 1 : 4
- (D) 2 : 3

84. CBR test is a type of

- (A) shear test
- (B) bearing test
- (C) penetration test
- (D) None of the above

85. Flash point test is a test for
- (A) soil
 - (B) aggregates
 - (C) bituminous material
 - (D) None of the above
86. The maximum spacing of expansion joints in plain concrete pavement 100 mm thick is
- (A) 27 m
 - (B) 37 m
 - (C) 47 m
 - (D) 57 m
87. Which of the following is **not** a type of rigid pavement failure?
- (A) Shrinkage cracks
 - (B) Mud pumping
 - (C) Warping cracks
 - (D) Ruts
88. Sheep foot roller is used for compacting
- (A) granular soil
 - (B) non-plastic silts
 - (C) fine sand
 - (D) clay soil
89. Dump trucks are
- (A) self-propelled hauling equipment
 - (B) towed hauling equipment
 - (C) self-propelled excavating equipment
 - (D) None of the above
90. Traffic rotary is suited for
- (A) T-intersection
 - (B) Y-intersection
 - (C) intersection
 - (D) staggered intersection
91. The minimum desirable radius of curve for a national highway in mountainous terrain, not snow bound, is
- (A) 50 m
 - (B) 40 m
 - (C) 30 m
 - (D) 15 m
92. Pick up the constituent of good brick earth whose excess causes the raw bricks shrink and warp during drying and burning, from the following.
- (A) Alumina
 - (B) Lime
 - (C) Iron oxide
 - (D) Magnesia

93. The difference between maximum time available and the actual time needed to perform an activity is known as
- (A) free float
 - (B) independent float
 - (C) total float
 - (D) half float
94. The simplest and most common method used in cities is to collect and dump waste in
- (A) landfill
 - (B) river
 - (C) sea
 - (D) None of the above
95. High turbidity of water can be determined by
- (A) turbidity tube
 - (B) Jackson turbidimeter
 - (C) Baylis turbidimeter
 - (D) Hellige turbidimeter
96. The minimum size of grains of silts is about
- (A) 0.0002 mm
 - (B) 0.002 mm
 - (C) 0.02 mm
 - (D) 0.2 mm
97. The distance between the centre of adjacent rivets in the same row is called
- (A) pitch
 - (B) lap
 - (C) gauge
 - (D) staggered lap
98. The main function of a spillway is to
- (A) discharge flood water
 - (B) supply water for irrigation
 - (C) supply water for power generation
 - (D) None of the above
99. Venturi meter is used to
- (A) measure the velocity of a flowing liquid
 - (B) measure the pressure of a flowing liquid
 - (C) measure the discharge of liquid flowing through a pipe
 - (D) measure the pressure difference of liquid flowing between two points in a pipeline
100. Which one is the correct sequence for the temporary adjustment of the theodolite?
- (A) Centering, elimination of parallax, levelling and setting
 - (B) Centering, setting, elimination of parallax and levelling
 - (C) Setting, centering, levelling and elimination of parallax
 - (D) Setting, levelling, elimination of parallax and centering

SPACE FOR ROUGH WORK

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SEAL

CEJ/DRW/I/25/7/7-A

16

26T—1500×4